

A Formally Verified Proof of Cramér's Theorem on Large Deviations

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What is Large Deviations Theory?

- Study of the **asymptotic decay of tail probabilities**
- Let $X_1, X_2, \dots$ be i.i.d. random variables with mean $\mu = \mathbb{E}[X_1]$
- By the **Law of Large Numbers**: $\bar{X}_n \rightarrow \mu$ a.s.
- By the **Central Limit Theorem**: $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$

Cramér's Theorem (1938) [5]

For $a \geq \mathbb{E}[X_1]$, the tail probability decays *exponentially*:

$$\frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) \rightarrow -I(a)$$

where $\Lambda(t) = \log \mathbb{E}[e^{tX_1}]$ is the **cumulant generating function** (CGF) and $I(a) = \sup_{t \in \mathbb{R}} \{ta - \Lambda(t)\}$ is the **rate function** (Legendre transform of the CGF).

Why Formal Verification?

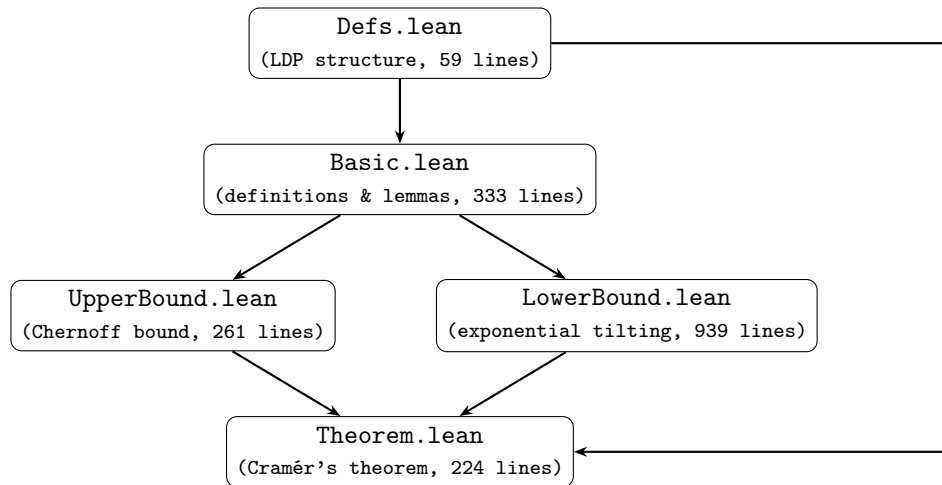
- Cramér's Theorem's proof involves delicate arguments often glossed over:
 - Limiting arguments in the lower bound proof
 - Coercions between $\mathbb{R}$ and $\overline{\mathbb{R}}$ (e.g., $\log 0 = -\infty$)
 - Precise enumeration of CLT preconditions under tilted measures
- Formal verification enforces **complete rigor**:
 - Every assumption explicitly enumerated in the Lean type signature
 - Machine-checked derivations — no hand-waved steps

Novel Contribution

First formalization of any large deviation result in a proof assistant (using Lean 4 / Mathlib)

- ① A formal definition of the **Large Deviation Principle** (new Lean structure)
- ② Proof of the **upper bound** via the Chernoff bound (building on Mathlib)
- ③ Proof of the **lower bound** via exponential tilting & change of measure
- ④ A precise **CLT assumption** under tilted measures, with informal justification
- ⑤ The **final LDP result** combining both bounds

Code Structure



Lower bound: $\approx$ **half** of the total codebase

Prior Formalizations of Probability

Isabelle/HOL

- Hölzl & Heller [11]: measure theory, Lebesgue integral, Radon-Nikodým, Fubini
- Avigad, Hölzl, Serafin [4]: **Central Limit Theorem**
- Hirata [10]: Lévy-Prokhorov metric

Rocq (Coq)

- Affeldt & Cohen [3]: measure & integration theory
- Affeldt et al. [2]: concentration inequalities, **Chernoff bound**

Lean 4 / Mathlib [13]

- Ying & Degenne [16]: Doob's martingale convergence
- Degenne [7]: Markov kernels
- Sonoda et al. [15]: Rademacher complexity
- SLLN (Etemadi's proof [9], in Mathlib)
- Marion, Zhu, Degenne [12]: **CLT** (merged March 2026)

Our Contribution in Context

- **No prior LDP formalization** in any proof assistant
 - Not Cramér's theorem, Sanov's theorem, nor Gärtner-Ellis
 - The LDP concept itself had not been formalized
- Closest prior work:
 - Chernoff bounds in Rocq [2] — used in our upper bound
 - CLT in Isabelle/HOL [4]; recently Lean/Mathlib [12]
- We introduce the LDP as a **formal Lean structure**, prove Cramér's theorem, and account for the CLT gap

Setup & Definitions

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $X_1, X_2, \dots$ be i.i.d. real-valued random variables.

Definition (Partial Sums and Empirical Mean)

$$S_n = \sum_{i=1}^n X_i, \quad \bar{X}_n = \frac{S_n}{n}$$

Definition (MGF, CGF, and Rate Function)

$$M_X(t) = \mathbb{E}[e^{tX_1}]$$

(Moment Generating Function)

$$\Lambda(t) = \log M_X(t) = \log \mathbb{E}[e^{tX_1}]$$

(Cumulant Generating Function)

$$I(x) = \sup_{t \in \mathbb{R}} \{tx - \Lambda(t)\}$$

(Rate Function / Legendre Transform of $\Lambda(t)$)

Since $t = 0$ yields $0 \cdot x - \Lambda(0) = 0$, we have $I(x) \geq 0$ for all x .

The Large Deviation Principle

Definition (LDP — General Form [8])

A sequence Z_n satisfies an **LDP** with rate function I if for all A :

$$-\inf_{x \in \overset{\circ}{A}} I(x) \leq \liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(Z_n \in A) \leq \limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(Z_n \in A) \leq -\inf_{x \in \bar{A}} I(x)$$

Specialization to upper tail sets $[a, \infty)$

Both bounds collapse to the same rate function $I(a)$:

$$\text{Upper: } \limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) \leq -I(a)$$

$$\text{Lower: } \liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) \geq -I(a)$$

Definition (Tilted Measure)

For a tilting parameter $t \in \mathbb{R}$, define $\mathbb{Q}_{n,t}$ via the Radon-Nikodym derivative:

$$\frac{d\mathbb{Q}_{n,t}}{d\mathbb{P}} = \frac{e^{tS_n}}{\mathbb{E}_{\mathbb{P}}[e^{tS_n}]} = \frac{e^{tS_n}}{(M_X(t))^n} = e^{tS_n - n\Lambda(t)}$$

We will show that:

- Reweights probability to make $\{\bar{X}_n \geq a\}$ a **typical event**
- Key tool for the **lower bound**: transforms rare event $\rightarrow$ typical event
- Under $\mathbb{Q}_{n,t}$: $\mathbb{E}_{\mathbb{Q}_{n,t}}[\bar{X}_n] = \Lambda'(t)$
- Pick t so that $\Lambda'(t) = a \implies$ mean shifts to a

Case $a < \mathbb{E}[X_1]$: Not a Rare Event

- When $a < \mathbb{E}[X_1]$, $\{\bar{X}_n \geq a\}$ is **not** a rare event
- By the **Strong Law of Large Numbers**:

$$\bar{X}_n \rightarrow \mathbb{E}[X_1] \text{ a.s.} \implies \mathbb{P}(\bar{X}_n \geq a) \rightarrow 1$$

- Define $I(a) = 0$ in this case; LDP bounds hold trivially:
 - Upper: $\mathbb{P}(\bar{X}_n \geq a) \leq 1 \implies \frac{1}{n} \log \mathbb{P} \leq 0 = -I(a)$
 - Lower: $\mathbb{P}(\bar{X}_n \geq a) \rightarrow 1 \implies \frac{1}{n} \log \mathbb{P} \rightarrow 0 = -I(a)$

Effective Rate Function

$$I(a) = \begin{cases} \sup_{t \in \mathbb{R}} \{ta - \Lambda(t)\} & \text{if } a \geq \mathbb{E}[X_1] \\ 0 & \text{if } a < \mathbb{E}[X_1] \end{cases}$$

- **Lean 4** [6]: programming language and proof assistant. Code that type-checks is provably correct.
- **Mathlib** [13]: community math library (>2M lines), providing Real Analysis, Measure Theory, and Probability Foundations: σ -algebras, probability measures, MGF/CGF, tilted measures, SLLN
- Built on **dependent type theory**, not set theory
 - E.g. $\mathbb{R}$ and $\overline{\mathbb{R}}$ ([EReal](#)) are distinct types:
 - $x \in \mathbb{R}$ is **not** implicitly an element of $\overline{\mathbb{R}}$
 - Requires explicit coercions $\uparrow x$, introduces overhead to formalization

- **Arguments:**
 - `{x}` – implicit (inferred by the type checker)
 - `(x)` – explicit (supplied by the caller)
 - `[instance]` – type class instance
- `noncomputable` – definition involves non-constructive axioms; not compilable to executable code, but fully type-checked and verified
- **Filters** – unified framework for limits and asymptotics:
 - $\forall^f n \text{ in } \text{atTop} = \text{“for all sufficiently large } n\text{”}$
 - `limsup` and `liminf` defined via filters
 - `Tendsto f atTop (N c) = “ $f(n) \rightarrow c$ ”`

Formal Definitions (Lean)

```
variable {Ω : Type*} [MeasureSpace Ω]
variable (X : ℕ → Ω → ℝ)

/-- The partial sum  $X_0 + \dots + X_{\{n-1\}}$ . -/
noncomputable def partialSum (n : ℕ) : Ω → ℝ :=  $\sum i \in \text{Finset.range } n, X\ i$ 

/-- The empirical mean  $S_n / n$ . -/
noncomputable def empiricalMean (n : ℕ) : Ω → ℝ := fun ω => partialSum X n ω / n

/-- Rate function: Legendre transform of the CGF. -/
noncomputable def I (x : ℝ) : ℝ :=  $\bigsqcup t : \mathbb{R}, t * x - \text{cgf } (X\ 0)\ \mathbb{P}\ t$ 

/-- Effective rate function for the LDP statement. -/
noncomputable def upperTailRateFunction (a : ℝ) : ℝ :=
  if  $\mathbb{E}[X\ 0] \leq a$  then I X a else 0

/-- The exponentially tilted measure, tilted by  $t \cdot S_n$ . -/
noncomputable def Qnt (μ : Measure Ω) (n : ℕ) (t : ℝ) :
  Measure Ω := Measure.tilted μ (fun ω => t * partialSum X n ω)
```

Note: $\bigsqcup$ is notation for the indexed supremum (`iSup`); Lean indexes from 0, so `X 0` corresponds to X_1

The Large Deviation Principle (Lean)

```
structure LargeDeviationPrinciple
  (Y : ℕ → Ω → ℝ) (I : ℝ → ℝ) : Prop where
  upperBound : ∀ a : ℝ,
    limsup (fun n : ℕ => ((1 : ℝ) / (n : ℝ) : EReal) * ENNReal.log (ℙ {ω | Y n ω ≥ a})) atTop
      ≤ (- I a : EReal)
  lowerBound : ∀ a : ℝ,
    (- I a : EReal) ≤
      liminf (fun n : ℕ => ((1 : ℝ) / (n : ℝ) : EReal) * ENNReal.log (ℙ {ω | Y n ω ≥ a})) atTop
```

- Uses `EReal` to handle $\log 0 = -\infty$ and `ENNReal.log`: $[0, \infty] \rightarrow [-\infty, \infty]$
- Separated upper/lower bounds for independent proofs

Assumptions (Part 1)

- **Measurability:** X_i are measurable

```
variable (h_meas : ∀ n, Measurable (X n))
```

- **Independence:** X_i are mutually independent

```
variable (h_indep : iIndepFun X ℙ)
```

- **Identically Distributed:** same distribution as X_0

```
variable (h_ident : ∀ n, IdentDistrib (X n) (X 0) ℙ ℙ)
```

- **Integrability:** X_1 measurable and $\mathbb{E}[|X_1|] < \infty$

```
variable (h_int : Integrable (X 0) ℙ)
```

- **Finite MGF:** $\mathbb{E}[e^{tX_1}] < \infty$ for all $t \in \mathbb{R}$

```
variable (h_mgf : ∀ t : ℝ,  
  Integrable (fun ω => Real.exp (t * X 0 ω)) ℙ)
```

Assumptions (Part 2)

- **Bounded rate function:** $\{ta - \Lambda(t)\}$ is bounded above

```
variable (h_bdd : ∀ a : ℝ,  
  BddAbove (Set.range (fun t => t * a - cgf (X 0) ℙ t)))
```

- **Non-degeneracy:** $\Lambda''(t) > 0$ for all t

```
variable (h_non_deg : ∀ t : ℝ,  
  0 < iteratedDeriv 2 (cgf (X 0) ℙ) t)
```

- **Exposed points:** $\forall a \geq \mathbb{E}[X], \exists t : \Lambda'(t) = a$

```
variable (h_exposed : ∀ a : ℝ,  $\mathbb{E}[X 0] \leq a \rightarrow$   
   $\exists t, \text{deriv } (\text{cgf } (X 0) \mathbb{P}) t = a$ )
```

Assumptions (Part 3): CLT under Tilted Measures

- **CLT under tilted measures:** For $\Lambda'(t) = a$, under the tilted measure $\mathbb{Q}_{n,t}$, the sample mean concentrates around a :

$$\forall \delta > 0, \forall \varepsilon > 0, \quad \mathbb{Q}_{n,t}(\overline{X}_n \in [a, a + \delta]) \geq \frac{1}{2} - \varepsilon \quad \text{eventually}$$

```
variable (h_clt_axiom :
  iIndepFun X  $\mathbb{P} \rightarrow$ 
  ( $\forall$  n, IdentDistrib (X n) (X 0)  $\mathbb{P}$   $\mathbb{P}$ )  $\rightarrow$ 
  ( $\forall$  n, Measurable (X n))  $\rightarrow$ 
  Integrable (X 0)  $\mathbb{P} \rightarrow$ 
  ( $\forall$  t :  $\mathbb{R}$ , Integrable (fun  $\omega \Rightarrow$  Real.exp (t * X 0  $\omega$ ))  $\mathbb{P}$ )  $\rightarrow$ 
  ( $\forall$  t :  $\mathbb{R}$ , 0 < iteratedDeriv 2 (cgf (X 0)  $\mathbb{P}$ ) t)  $\rightarrow$ 
   $\forall$  (t a :  $\mathbb{R}$ ),  $\forall$   $\delta > 0$ ,  $\forall$   $\varepsilon > 0$ ,
  deriv (cgf (X 0)  $\mathbb{P}$ ) t = a  $\rightarrow$ 
     $\forall^f$  n in atTop,
    (1 / 2 -  $\varepsilon$  :  $\mathbb{R}$ )  $\leq$ 
      ((Measure.tilted  $\mathbb{P}$ 
        (fun  $\omega \Rightarrow$  t * ( $\sum$  i  $\in$  Finset.range n, X i  $\omega$ )))
        { $\omega$  | ( $\sum$  i  $\in$  Finset.range n, X i  $\omega$ ) / n
           $\in$  Set.Icc a (a +  $\delta$ )}.toReal)
```

Cramér's Theorem — Formal Statement

Theorem (Cramér's Theorem)

The empirical means $\bar{X}_n$ satisfy an LDP with rate function *upperTailRateFunction*:

- For $a \geq \mathbb{E}[X]$: rate function $I(a) = \sup_t \{ta - \Lambda(t)\}$
- For $a < \mathbb{E}[X]$: rate function 0 (trivial case via SLLN)

```
include h_indep h_meas h_ident h_mgf h_int h_bdd h_non_deg h_exposed h_clt_axiom in
/-- **Cramér's Theorem**: For i.i.d. random variables
with finite MGF, the empirical mean satisfies a large
deviation principle with rate function given by the
Legendre transform of the CGF. -/
theorem cramers_theorem : LargeDeviationPrinciple (empiricalMean X) (upperTailRateFunction X)
```

Verified by Lean's type-theoretic kernel

- ① Finite MGF of identically distributed variables: $\mathbb{E}[e^{tX_i}] < \infty$ for all i, t

```
lemma integrable_exp_of_identDistrib (i : ℕ) (t : ℝ) : Integrable (fun ω => Real.exp (t * X i ω)) ℙ
```

- ② Measurability of partial sum S_n :

```
lemma measurable_partialSum (n : ℕ) : Measurable (partialSum X n)
```

- ③ Measurability of empirical mean $\overline{X}_n$:

```
lemma measurable_empiricalMean (n : ℕ) : Measurable (empiricalMean X n)
```

- ④ Interior of integrable exponential set: $\forall t \in \mathbb{R}, t \in \text{int}\{t \mid \mathbb{E}[e^{tX_1}] < \infty\}$

```
lemma mem_interior_integrableExpSet (t : ℝ) :  
  t ∈ interior (integrableExpSet (X 0) ℙ)
```

Basic Lemmas (Part 2)

5 CGF lower bound: $\Lambda(t) \geq t\mathbb{E}[X]$ (Jensen's inequality)

```
lemma cgf_ge_mul_expect {Y :  $\Omega \rightarrow \mathbb{R}$ } (h_int : Integrable Y  $\mathbb{P}$ )  
  (h_mgf :  $\forall t : \mathbb{R}, \text{Integrable } (\text{fun } \omega \Rightarrow \text{Real.exp } (t * Y \omega)) \mathbb{P}$ ) (t :  $\mathbb{R}$ ) :  
  cgf Y  $\mathbb{P}$  t  $\geq t * \mathbb{E}[Y]$ 
```

6 Rate function negative for negative parameter: if $t < 0$ and $a \geq \mathbb{E}[X]$: $ta - \Lambda(t) \leq 0$

```
lemma rate_function_neg_param_le_zero {Y :  $\Omega \rightarrow \mathbb{R}$ } (h_int : Integrable Y  $\mathbb{P}$ )  
  (h_mgf :  $\forall t : \mathbb{R}, \text{Integrable } (\text{fun } \omega \Rightarrow \text{Real.exp } (t * Y \omega)) \mathbb{P}$ )  
  {t a :  $\mathbb{R}$ } (ht : t < 0) (ha :  $\mathbb{E}[Y] \leq a$ ) : t * a - cgf Y  $\mathbb{P}$  t  $\leq 0$ 
```

7 Finite MGF of partial sum S_n :

```
lemma integrable_exp_sum (t :  $\mathbb{R}$ ) (n :  $\mathbb{N}$ ) :  
  Integrable (fun  $\omega \Rightarrow \text{Real.exp } (t * \text{partialSum } X \ n \ \omega)) \mathbb{P}$ 
```

8 Tilted measure is a probability measure: $\mathbb{Q}_{n,t}(\Omega) = 1$

```
lemma isProbabilityMeasure_tilted_partialSum (t :  $\mathbb{R}$ ) (n :  $\mathbb{N}$ ) : IsProbabilityMeasure ( $\mathbb{Q}_{n,t}$  X  $\mathbb{P}$  n t)
```

Basic Lemmas (Part 3)

9 Interior of integrable exp. set of partial sums:

```
lemma mem_interior_integrableExpSet_partialSum (t : ℝ) (n : ℕ) :  
  t ∈ interior (integrableExpSet (partialSum X n) ℙ)
```

10 Rate function supremum at non-negative t : $I(a) = \sup_{t \geq 0} \{ta - \Lambda(t)\}$ for $a \geq \mathbb{E}[X]$

```
lemma rateFunction_eq_sup_nonneg (a : ℝ) (h_mean :  $\mathbb{E}[X] \leq a$ ) :  
  I X a =  $\bigvee t : \{(x : \mathbb{R}) \mid 0 \leq x\}, (t : \mathbb{R}) * a - \text{cgf } (X) \mathbb{P} t$ 
```

11 MGF of partial sum: $M_{S_n}(t) = e^{n\Lambda(t)}$

```
lemma mgf_sum_eq_exp_n_prod_cgf (n : ℕ) (t : ℝ) :  
  mgf (partialSum X n) ℙ t = Real.exp (n * cgf (X) ℙ t)
```

12 CGF of partial sum: $\Lambda_{S_n}(t) = n\Lambda(t)$

```
lemma cgf_sum_eq_n_prod_cgf (n : ℕ) (t : ℝ) :  
  cgf (partialSum X n) ℙ t = (n : ℝ) * cgf (X) ℙ t
```

Goal

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) \leq -I(a)$$

Approach: Exponential Markov's inequality (= Chernoff bound)

- 1 Apply Markov's inequality to e^{tS_n}
- 2 Factor the expectation using independence
- 3 Take logarithms and optimize over $t \geq 0$
- 4 Show the supremum over $t \geq 0$ equals $I(a)$

Chernoff Bound Derivation

Since $x \mapsto e^{tx}$ is monotonically increasing, apply Markov's inequality:

$$\mathbb{P}(\bar{X}_n \geq a) = \mathbb{P}(e^{tS_n} \geq e^{tna}) \leq e^{-tna} \mathbb{E}[e^{tS_n}]$$

By independence, the expectation factors:

$$\mathbb{E}[e^{tS_n}] = \prod_{i=1}^n \mathbb{E}[e^{tX_i}] = \left(\mathbb{E}[e^{tX_1}] \right)^n = e^{n\Lambda(t)}$$

Therefore:

$$\boxed{\mathbb{P}(\bar{X}_n \geq a) \leq e^{-n(ta - \Lambda(t))}}$$

From Chernoff to Rate Function

Taking $\frac{1}{n} \log$:

$$\frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) \leq -(ta - \Lambda(t))$$

The bound is **independent of** n , so taking $\limsup$ preserves it.

Optimizing over $t \geq 0$:

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) \leq -\sup_{t \geq 0} \{ta - \Lambda(t)\}$$

Key identity

$$\sup_{t \geq 0} \{ta - \Lambda(t)\} = \sup_{t \in \mathbb{R}} \{ta - \Lambda(t)\} = I(a)$$

Proof: If $t < 0$ and $a \geq \mathbb{E}[X]$, then $ta - \Lambda(t) \leq 0$ by Jensen's inequality ($\Lambda(t) \geq t\mathbb{E}[X]$). Since $t \cdot 0 - \Lambda(0) = 0$ negative t cannot improve the supremum.

Worked Example: Exponential Distribution

Let $X_i \sim \text{Exp}(\lambda)$ with $\mathbb{E}[X_1] = 1/\lambda$.

CGF:

$$\Lambda(t) = \log \lambda - \log(\lambda - t), \quad t < \lambda$$

Rate function: Optimize $g(t) = ta - \Lambda(t)$:

$$g'(t) = 0 \implies t^* = \lambda - \frac{1}{a}$$

$$I(a) = \lambda a - 1 - \log(\lambda a)$$

Sanity check:

- At $a = 1/\lambda$ (the mean):

$$I(1/\lambda) = 1 - 1 - \log 1 = 0 \quad \checkmark$$

- As a increases: $I(a) > 0$
- Exponential decay of tail confirmed

Upper bound:

$$\limsup \frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) \leq -I(a)$$

Formalized Upper Bound (Lean)

Chernoff bound lemma:

```
lemma prob_mean_ge_le_exp (t a : ℝ) (ht : 0 ≤ t)
  (n : ℕ) (hn_pos : 0 < n) :
  (ℙ {ω | empiricalMean X n ω ≥ a}).toReal
    ≤ Real.exp (-(n : ℝ) * (t * a - cgf (X 0) ℙ t))
```

Upper bound theorem:

```
theorem cramer_upper_bound (a : ℝ) (h_mean : E[X 0] ≤ a) :
  limsup (fun n : ℕ =>
    ((1 : ℝ) / (n : ℝ) : EReal) *
      ENNReal.log (ℙ {ω | empiricalMean X n ω ≥ a}))
    atTop ≤ (- I X a : EReal)
```

- Several **EReal coercion lemmas** needed to lift $\mathbb{R}$ results to $\overline{\mathbb{R}}$
- Infimum of negations, coercion of infima, etc.

Goal

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) \geq -I(a)$$

Approach: Change of measure to tilted $\mathbb{Q}_{n,t}$

- 1 Restrict to a concentrated set $E_n = \{\bar{X}_n \in [a, a + \delta]\}$
- 2 Change of measure: bound $\mathbb{P}(E_n)$ from below using $\mathbb{Q}_{n,t}(E_n)$
- 3 Use CLT to show $\mathbb{Q}_{n,t}(E_n)$ is bounded away from 0
- 4 Take limits and send $\delta \rightarrow 0$

Key idea: Tilting transforms the rare event into a typical one

Why Tilting Works

Rearranging the Radon-Nikodym derivative:

$$d\mathbb{P} = e^{-n(t\bar{X}_n - \Lambda(t))} d\mathbb{Q}_{n,t}$$

This **matches** the Chernoff bound $\mathbb{P}(\bar{X}_n \geq a) \leq e^{-n(ta - \Lambda(t))}$, except $\bar{X}_n$ replaces a .

If we could show $\bar{X}_n \approx a$ under $\mathbb{Q}_{n,t}$, the bound becomes tight!

Mean and Variance under $\mathbb{Q}_{n,t}$

$$\mathbb{E}_{\mathbb{Q}_{n,t}}[\bar{X}_n] = \Lambda'(t) \quad (\text{pick } t \text{ so } \Lambda'(t) = a)$$

$$\mathbb{V}_{\mathbb{Q}_{n,t}}(\bar{X}_n) = \frac{\Lambda''(t)}{n} \rightarrow 0$$

Tilted measure **re-centers** at a with **shrinking variance**

CLT under Tilted Measures

Define standardized variables:

$$Y_i = \frac{X_i - \Lambda'(t)}{\sqrt{\Lambda''(t)}}, \quad Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i$$

Key computation: The characteristic function of Y_i under $\mathbb{Q}_{n,t}$ is

$$\varphi_{Y_i, \mathbb{Q}_{n,t}}(\theta) = \boxed{e^{-\Lambda(t)} \mathbb{E}_{\mathbb{P}}[e^{i\theta Y_i + tX_i}]}$$

which is **independent of n** . Therefore:

$$\varphi_{Z_n, \mathbb{Q}_{n,t}}(\theta) = \left(\varphi_{Y_1, \mathbb{Q}_{n,t}}\left(\frac{\theta}{\sqrt{n}}\right) \right)^n \xrightarrow{n \rightarrow \infty} e^{-\theta^2/2}$$

By **Lévy's Continuity Theorem**: $Z_n \xrightarrow{d} \mathcal{N}(0, 1)$ under $\mathbb{Q}_{n,t}$.

CLT Corollary (the assumption)

Corollary

For any $\delta > 0$:

$$\mathbb{Q}_{n,t}(\bar{X}_n \in [a, a + \delta]) \rightarrow \frac{1}{2}$$

Proof sketch: The event $\bar{X}_n \in [a, a + \delta]$ is equivalent to $Z_n \in [0, \delta\sqrt{n/\Lambda''(t)}]$. As the upper bound $\rightarrow \infty$, the probability converges to $1/2$ by the standard normal CDF: $\Phi(\infty) - \Phi(0) = 1/2$.

Formalized as [h_clt_axiom](#):

- All preconditions explicitly listed for future replacement
- Once Mathlib CLT is adapted to tilted measures, this axiom can be directly replaced

Lower Bound Proof

Restrict to $E_n = \{\bar{X}_n \in [a, a + \delta]\}$.

Step 1: Change of measure

$$\mathbb{P}(\bar{X}_n \geq a) \geq \mathbb{P}(E_n) \geq e^{-n(t(a+\delta) - \Lambda(t))} \mathbb{Q}_{n,t}(E_n)$$

Step 2: Take scaled logarithms

$$\frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) \geq -(ta - \Lambda(t)) - t\delta + \frac{1}{n} \log \mathbb{Q}_{n,t}(E_n)$$

Step 3: CLT gives $\mathbb{Q}_{n,t}(E_n) \geq c > 0$, so $\frac{1}{n} \log \mathbb{Q}_{n,t}(E_n) \rightarrow 0$

Step 4: Send $\delta \rightarrow 0$:

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) \geq -(ta - \Lambda(t))$$

Step 5: Identify $I(a) = ta - \Lambda(t)$ (unique maximizer since Λ strictly convex)

Worked Example: Exponential Distribution (Lower Bound)

Tilted distribution: Under $\mathbb{Q}_{n,t^*}$ with $t^* = \lambda - 1/a$:

$$e^{t^*x} \cdot \lambda e^{-\lambda x} \propto e^{-x/a} \implies X_i \sim \text{Exp}(1/a)$$

The exponential family is **closed under tilting**.

Shifted mean: $\mathbb{E}_{\mathbb{Q}_{n,t^*}}[X_i] = a$ — rare event becomes typical!

Lower bound:

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) \geq -I(a) = -(\lambda a - 1 - \log(\lambda a))$$

Matching the upper bound $\implies$ the LDP is tight:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) = -I(a) \quad \checkmark$$

Formalized Lower Bound (Lean)

Change of measure: For $E = \{\bar{X}_n \in [a, a + \delta]\}$ and $t > 0$: $\mathbb{P}(E) \geq e^{-n(t(a+\delta) - \Lambda(t))} \mathbb{Q}_{n,t}(E)$

```
lemma change_of_measure_lower_bound (a δ t : ℝ)
  (n : ℕ) (_hδ : 0 < δ) (ht : 0 < t) ... :
  let E := {ω | empiricalMean X n ω ∈ Set.Icc a (a + δ)}
  (ℙ E).toReal ≥
    Real.exp (-n * (t * (a + δ) - cgf (X 0) ℙ t)) * ((ℚnt X ℙ n t) E).toReal
```

Lower bound theorem: For $a \geq \mathbb{E}[X]$: $\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) \geq -I(a)$

```
theorem cramer_lower_bound (a : ℝ) (h_mean : ℙ[X 0] ≤ a) :
  (- I X a : EReal) ≤
    liminf (fun n : ℕ => ((1 : ℝ) / (n : ℝ) : EReal)
      * ENNReal.log (ℙ {ω | empiricalMean X n ω ≥ a})) atTop
```

Lower bound code is the **plurality** of the codebase ($\approx$ half)

Putting It All Together

theorem `cramers_theorem` :

`LargeDeviationPrinciple` (`empiricalMean` `X`) (`upperTailRateFunction` `X`)

Case 1: $a \geq \mathbb{E}[X_1]$

- **Upper bound** (Chernoff): $\limsup \frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) \leq -I(a)$
- **Lower bound** (tilting): $\liminf \frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) \geq -I(a)$

Combined: $\lim \frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) = -I(a)$

Case 2: $a < \mathbb{E}[X_1]$

Rate function $I(a) = 0$ by definition of `upperTailRateFunction`.

- **Upper:** $\mathbb{P} \leq 1 \implies \frac{1}{n} \log \mathbb{P} \leq 0 = -I(a)$
- **Lower:** SLLN gives $\mathbb{P}(\bar{X}_n \geq a) \rightarrow 1$, so $\frac{1}{n} \log \mathbb{P} \rightarrow 0$

Formalization overhead

- Type coercion is an example: LDP uses $\overline{\mathbb{R}}$ ([EReal](#)); intermediate computations use $\mathbb{R}$
- $\mathbb{R} \subset \overline{\mathbb{R}}$ is taken for granted in set theory; in Lean, requires explicit coercion lemmas

Lower bound complexity

- $\approx$ half the codebase, despite comparable informal proof length
- Change-of-measure, sequence of tilted measures, formal bookkeeping
- “Routine” steps require explicit justification in Lean

Filter-based asymptotics

- Mathlib unifies limits, convergence, “for all sufficiently large n ”
- Elegant but requires adjustment from standard ε - N practice

Built on existing foundations:

- Measure theory, Chernoff bounds, tilted measures
- Strong Law of Large Numbers, extended reals
- Analysis: Jensen's inequality, convexity, derivatives

Contributed back:

- 2 general-purpose CGF lemmas to Mathlib's Moments module
- Various intermediate results applicable to future work

Rapid growth of Mathlib probability:

- Martingale convergence, Markov kernels, sub-Gaussian variables
- CLT recently merged (Marion, Zhu, Degenne – March 2026)
- Foundation for further advanced probability formalization

Removing the CLT Axiom

- **Mathlib CLT** recently merged (Marion, Zhu, Degenne 2026)
 - Uses characteristic functions + Lévy's Continuity Theorem
 - Same approach as our informal proof of `h_clt_axiom`
- **Adaptation needed** for tilted-measure setting:
 - Standard CLT: fixed measure, i.i.d. variables
 - Our setting: sequence of measures $\mathbb{Q}_{n,t}$
 - Key insight: characteristic function of Y_i under $\mathbb{Q}_{n,t}$ is independent of n
- `h_clt_axiom` was **structured for replacement**:
 - All preconditions explicitly enumerated
 - Can be swapped for a derived result directly

Generalizations

General LDP for arbitrary sets:

- Over topological spaces $\mathcal{X}$ with arbitrary Borel sets B :

$$-\inf_{x \in \overset{\circ}{B}} I(x) \leq \liminf_{\epsilon \rightarrow 0} \epsilon \log \mu_\epsilon(B) \leq \limsup_{\epsilon \rightarrow 0} \epsilon \log \mu_\epsilon(B) \leq -\inf_{x \in \overline{B}} I(x)$$

Relaxing assumptions:

- MGF finiteness — many distributions have partial domains
- Exposed point condition — currently excludes distributions like Bernoulli

Extensions:

- $\mathbb{R}^d$ Cramér's theorem (multidimensional CGF)
- Gärtner-Ellis theorem (relaxes i.i.d. assumption)
- Sanov's theorem (empirical measures, KL divergence)
- Contraction principle (deriving new LDPs from existing ones)

- LLMs increasingly adopted in the Lean community:
 - **Lean Copilot** [14]: in-editor tactic suggestions
 - **Aristotle** [1]: LLM reasoning + formal proof search
- In this project, **Claude Code** assisted with:
 - Lean 4 coding and Mathlib API navigation
 - Discovering proofs and strategies
 - Debugging type errors
- **Vision for the future:**
 - Unlike most LLM output, Lean proofs are **mechanically checkable**
 - LLMs as directed search over derivation space
 - Human explores theory $\longrightarrow$ LLM formalizes and verifies

- **First formalization of a Large Deviation result** in any proof assistant
- **1,819 lines** of Lean 4, building on Mathlib
- Formal LDP definition, upper bound (Chernoff), lower bound (exponential tilting)
- CLT assumption with informal proof; structured for future replacement



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Thank you!

Questions?

Appendix: Upper Bound — Intermediate Lemmas

❶ **Infimum of negation:** For $S \subseteq \mathbb{R}$ bounded above, $\inf\{-x \mid x \in S\} = -\sup S$

```
private lemma csInf_neg_eq_neg_csSup {ι : Type*} (f : ι → ℝ)
  (hne : (Set.range f).Nonempty) (hbdd_above : BddAbove (Set.range f)) :
  sInf (Set.range fun i => -f i) = -sSup (Set.range f)
```

❷ **Infimum coercion to $\overline{\mathbb{R}}$:** For $S \subseteq \mathbb{R}$ bounded below, $\inf S_\infty = (\inf S)_\infty$

```
private lemma ereal_sInf_coe_eq_coe_sInf {s : Set ℝ}
  (hne : s.Nonempty) (hbdd : BddBelow s) :
  sInf ((fun (x : ℝ) => (x : EReal)) '' s) = ↑(sInf s)
```

❸ **Infimum of negations in $\overline{\mathbb{R}}$:** For $S \subseteq \mathbb{R}$ bounded above, $\inf\{-x_\infty \mid x \in S\} = -(\sup S)_\infty$

```
private lemma ereal_sInf_neg_eq_neg_sSup {ι : Type*} (f : ι → ℝ)
  (hne : (Set.range f).Nonempty) (hbdd : BddAbove (Set.range f)) :
  sInf (Set.range fun i => -(f i)) = -((sSup (Set.range f) : ℝ) : EReal)
```

❹ **Chernoff bound for empirical mean:** For $t \geq 0$, $\mathbb{P}(\overline{X}_n \geq a) \leq e^{-n(ta - \Lambda(t))}$

```
lemma prob_mean_ge_le_exp (t a : ℝ) (ht : 0 ≤ t) (n : ℕ) (hn_pos : 0 < n) :
  (ℙ {ω | empiricalMean X n ω ≥ a}).toReal ≤ Real.exp (-(n : ℝ) * (t * a - cgf (X 0) ℙ t))
```

❺ **Cramér upper bound:** $\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(\overline{X}_n \geq a) \leq -I(a)$

```
theorem cramer_upper_bound (a : ℝ) (h_mean : E[X 0] ≤ a) :
  limsup (fun n : ℕ => ((1 : ℝ) / (n : ℝ) : EReal) * ENNReal.log (ℙ {ω | empiricalMean X n ω ≥ a})) atTop
  ≤ (-I X a : EReal)
```

Appendix: Lower Bound — Intermediate Lemmas (1/7)

- ❶ **Exponential bound on concentrated set:** For $t \geq 0$, if $\bar{X}_n \in [a, a + \delta]$, then
- $$e^{-tS_n} \geq e^{-tn(a+\delta)}$$

```
private lemma exp_neg_mul_S_ge_on_set (t : ℝ) (n : ℕ) (a δ : ℝ) (ht : 0 ≤ t)
  (ω : Ω) (hω : empiricalMean X n ω ∈ Set.Icc a (a + δ)) :
  Real.exp (-t * partialSum X n ω) ≥ Real.exp (-t * n * (a + δ))
```

- ❷ **Tilted window probability lower bound:** Given $\Lambda'(t) = a$, $\exists c > 0$ s.t. eventually
- $$\mathbb{Q}_{n,t}(\bar{X}_n \in [a, a + \delta]) \geq c$$

```
private lemma tilted_window_lower_bound_from_concentration (a t δ : ℝ) (hδ : 0 < δ)
  (ht_deriv : deriv (cgf (X 0) ℙ) t = a) :
  ∃ c > 0, ∀f n in atTop,
  c ≤ ((Qnt X ℙ n t) {ω | empiricalMean X n ω ∈ Set.Icc a (a + δ)}).toReal
```

- ❸ **Vanishing scaled constant in $\overline{\mathbb{R}}$:** $\lim_{n \rightarrow \infty} \frac{c}{n} = 0$

```
private lemma ereal_inv_nat_mul_const_tendsto_zero (c : ℝ) :
  Tendsto (fun n : ℕ => ((1 : ℝ) / n : EReal) * (c : EReal)) atTop (N 0)
```

- ❹ **Log product splitting:** $\frac{1}{n} \log(e^x \cdot y) = \frac{x}{n} + \frac{1}{n} \log y$ in $\overline{\mathbb{R}}$

```
private lemma log_product_split (n : ℕ) (x : ℝ) (y : ENNReal) (hn : n ≥ 1)
  (hy_ne_zero : y ≠ 0) (hy_ne_top : y ≠ ⊤) :
  ((1 : ℝ) / n : EReal) * ENNReal.log (ENNReal.ofReal (Real.exp x * y.toReal)) =
  ((1 : ℝ) / n : EReal) * (x : EReal) + ((1 : ℝ) / n : EReal) * ENNReal.log y
```

Appendix: Lower Bound — Intermediate Lemmas (2/7)

5 Simplified logarithmic expansion: Algebraic simplification of the log term in the proof

```
private lemma log_exp_product_eq_neg_coef_plus_log (n : ℕ) (t a δ : ℝ)
  (cgf_val : ℝ) (μ : ENNReal) (hn : n ≥ 1)
  (hμ_ne_zero : μ ≠ 0) (hμ_ne_top : μ ≠ 1) :
  ((1 : ℝ) / n : EReal) * ENNReal.log (ENNReal.ofReal
    (Real.exp (-n * (t * (a + δ) - cgf_val)) * μ.toReal))) =
  (-(t * a - cgf_val) - t * δ : EReal) + ((1 : ℝ) / n : EReal) * ENNReal.log μ
```

6 Constant plus vanishing sequence: If $f_n \rightarrow 0$, then $c + f_n \rightarrow c$ in $\overline{\mathbb{R}}$

```
private lemma tendsto_const_add_vanishing (c : EReal) (f : ℕ → EReal)
  (h : Tendsto f atTop (ℕ 0)) : Tendsto (fun n => c + f n) atTop (ℕ c)
```

7 EReal limit bound: If $x - \varepsilon \leq y$ for all $\varepsilon > 0$, then $x \leq y$ in $\overline{\mathbb{R}}$

```
private lemma EReal.le_of_forall_pos_sub_le {x y : EReal}
  (h : ∀ ε : ℝ, 0 < ε → x - (ε : EReal) ≤ y) : x ≤ y
```

8 Measure via tilted integral: $\mathbb{P}(E) = \mathbb{E}[e^f] \int_E e^{-f} d\mathbb{Q}$

```
private lemma measure_eq_integral_exp_neg_tilted (f : Ω → ℝ) (E : Set Ω)
  (h_int : Integrable (fun ω => Real.exp (f ω)) ℙ) (hE : MeasurableSet E) :
  (ℙ E).toReal =
  (ℙ E) *
  (∫ ω in E, Real.exp (-f ω) ∂(Measure.tilted ℙ f))
```

Appendix: Lower Bound — Intermediate Lemmas (3/7)

9 Change of measure lower bound: $\mathbb{P}(E) \geq e^{-n(t(a+\delta)-\Lambda(t))} \mathbb{Q}_{n,t}(E)$

```
lemma change_of_measure_lower_bound (a δ t : ℝ) (n : ℕ) (hδ : 0 < δ) (ht : 0 < t)
(h_int : Integrable (fun ω => Real.exp (t * partialSum X n ω)) P) :
  let E := {ω | empiricalMean X n ω ∈ Set.Icc a (a + δ)}
  (P E).toReal ≥
    Real.exp (-n * (t * (a + δ) - cgf (X 0) P t)) * ((Qnt X P n t) E).toReal
```

10 Error term vanishes: $\frac{1}{n} \log \mathbb{Q}_{n,t}(\overline{X}_n \in [a, a + \delta]) \rightarrow 0$

```
private lemma error_term_vanishes (a t δ : ℝ) (hδ : 0 < δ)
(ht_int : t ∈ interior (integrableExpSet (X 0) P))
(ht_deriv : deriv (cgf (X 0) P) t = a) :
  Tendsto (fun n : ℕ => ((1 : ℝ) / n : EReal) * ENNReal.log ((Qnt X P n t)
    {ω | empiricalMean X n ω ∈ Set.Icc a (a + δ)})) atTop (N 0)
```

11 Target bound for lower limit: $\liminf \frac{1}{n} \log \mathbb{P}(\overline{X}_n \geq a) \geq -(ta - \Lambda(t)) - t\delta$

```
private lemma lower_bound_via_tilted (a t δ : ℝ) (hδ : 0 < δ) (ht : 0 < t)
(ht_int : t ∈ interior (integrableExpSet (X 0) P))
(ht_deriv : deriv (cgf (X 0) P) t = a) :
  liminf (fun n : ℕ => ((1 : ℝ) / n : EReal) *
    ENNReal.log (P {ω | empiricalMean X n ω ≥ a})) atTop
  ≥ (-(t * a - cgf (X 0) P t) : EReal) - (t * δ : EReal)
```

Appendix: Lower Bound — Intermediate Lemmas (4/7)

- 12 **Derivative non-negativity:** If $a \geq \mathbb{E}[X]$ and $\Lambda'(t) = a$, then $t \geq 0$

```
private lemma deriv_cgf_nonneg_of_ge_mean (a : ℝ) (h_mean :  $\mathbb{E}[X \ 0] \leq a$ )  
  (t : ℝ) (ht_deriv : deriv (cgf (X 0)  $\mathbb{P}$ ) t = a) : 0 ≤ t
```

- 13 **Legendre transform maximum:** $I(a) = ta - \Lambda(t)$ when $\Lambda'(t) = a$

```
private lemma rateFunction_eq_of_deriv_eq (a t : ℝ)  
  (ht_deriv : deriv (cgf (X 0)  $\mathbb{P}$ ) t = a) : I X a = t * a - cgf (X 0)  $\mathbb{P}$  t
```

- 14 **Cramér lower bound:** $\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(\bar{X}_n \geq a) \geq -I(a)$

```
theorem cramer_lower_bound (a : ℝ) (h_mean :  $\mathbb{E}[X \ 0] \leq a$ ) :  
  ( $\neg$  I X a : EReal) ≤ liminf (fun n : ℕ => ((1 : ℝ) / (n : ℝ) : EReal) *  
    ENNReal.log ( $\mathbb{P}$  { $\omega$  | empiricalMean X n  $\omega$  ≥ a})) atTop
```

Appendix: Lower Bound — Supplementary Lemmas (5/7)

The following are supplementary lemmas towards formally verifying the CLT assumption.

15 **Vanishing variance term:** $\lim_{n \rightarrow \infty} \frac{1}{n} \frac{C}{\delta^2} = 0$

```
private lemma variance_term_tendsto_zero (C : ℝ) (δ : ℝ) (hδ : 0 < δ) :  
  Tendsto (fun n : ℕ => ENNReal.ofReal ((1 / n) * C / δ ^ 2)) atTop (N 0)
```

16 **Equality from integral identity:** If $f \leq c$ a.e. and $\int f d\mu = c$, then $f = c$ a.e.

```
private lemma ae_eq_const_of_integral_eq_const_of_le (μ : Measure α)  
  [IsProbabilityMeasure μ] {f : α → ℝ} (c : ℝ) (hf : Integrable f μ)  
  (h_le : ∀m ω ∂μ, f ω ≤ c) (h_int : μ[f] = c) : ∀m ω ∂μ, f ω = c
```

17 **CGF derivative for sum:** $\Lambda'_{S_n}(t) = n\Lambda'(t)$

```
private lemma deriv_cgf_sum (t : ℝ) (n : ℕ)  
  (ht : t ∈ interior (integrableExpSet (X 0) ℙ)) :  
  deriv (cgf (partialSum X n) ℙ) t = n * deriv (cgf (X 0) ℙ) t
```

18 **CGF second derivative for sum:** $\Lambda''_{S_n}(t) = n\Lambda''(t)$

```
private lemma iteratedDeriv_two_cgf_sum (t : ℝ) (n : ℕ)  
  (ht : t ∈ interior (integrableExpSet (X 0) ℙ)) :  
  iteratedDeriv 2 (cgf (partialSum X n) ℙ) t =  
    n * iteratedDeriv 2 (cgf (X 0) ℙ) t
```

Appendix: Lower Bound — Supplementary Lemmas (6/7)

19 Moments of S_n under tilting: $\mathbb{E}_{\mathbb{Q}}[S_n] = \Lambda'_{S_n}(t)$, $\mathbb{V}_{\mathbb{Q}}(S_n) = \Lambda''_{S_n}(t)$

```
private lemma tilted_Sn_moments (t : ℝ) (n : ℕ)
  (_ht : t ∈ interior (integrableExpSet (X 0) ℙ)) :
  let μ_t := Measure.tilted ℙ (fun ω => t * partialSum X n ω)
  μ_t[partialSum X n] = deriv (cgf (partialSum X n) ℙ) t ∧
  variance (partialSum X n) μ_t = iteratedDeriv 2 (cgf (partialSum X n) ℙ) t
```

20 Moments of $\bar{X}_n$ under tilting: $\mathbb{E}_{\mathbb{Q}}[\bar{X}_n] = \Lambda'(t)$, $\mathbb{V}_{\mathbb{Q}}(\bar{X}_n) = \frac{1}{n}\Lambda''(t)$

```
private lemma tilted_empirical_moments (t : ℝ) (n : ℕ) (hn : n ≠ 0)
  (ht : t ∈ interior (integrableExpSet (X 0) ℙ)) :
  let μ_t := Measure.tilted ℙ (fun ω => t * partialSum X n ω)
  μ_t[empiricalMean X n] = deriv (cgf (X 0) ℙ) t ∧
  variance (empiricalMean X n) μ_t = (1 / n) * iteratedDeriv 2 (cgf (X 0) ℙ) t
```

21 Chebyshev deviation bound: $\mathbb{Q}_{n,t}(|\bar{X}_n - a| \geq \delta) \leq \frac{1}{n} \frac{\Lambda''(t)}{\delta^2}$

```
private lemma tilted_deviation_bound (t a : ℝ) (n : ℕ) (hn : n ≠ 0)
  (δ : ℝ) (hδ : 0 < δ) (ht : t ∈ interior (integrableExpSet (X 0) ℙ))
  (h_match : deriv (cgf (X 0) ℙ) t = a) :
  let μ_t := Measure.tilted ℙ (fun ω => t * partialSum X n ω)
  μ_t {ω | δ ≤ |empiricalMean X n ω - a|} ≤
    ENNReal.ofReal ((1 / n) * iteratedDeriv 2 (cgf (X 0) ℙ) t / δ ^ 2)
```


22 **Window concentration:** $\mathbb{Q}_{n,t}(|\overline{X}_n - a| < \delta) \rightarrow 1$ as $n \rightarrow \infty$

```
private lemma tilted_measure_concentrates (t a δ : ℝ) (hδ : 0 < δ)
  (ht : t ∈ interior (integrableExpSet (X 0) ℙ))
  (h_match : deriv (cgf (X 0) ℙ) t = a) :
  Tendsto (fun n => ((Qnt X ℙ n t)
    {ω | |empiricalMean X n ω - a| < δ}).toReal) atTop (N 1)
```

23 **Tilted measure mean support:** $\mathbb{Q}_{n,t}(\overline{X}_n \geq a) > 0$

```
private lemma tilted_prob_ge_mean_pos (a t : ℝ)
  (ht_int : t ∈ interior (integrableExpSet (X 0) ℙ))
  (ht_deriv : deriv (cgf (X 0) ℙ) t = a) (n : ℕ) (hn : n ≠ 0) :
  0 < (Qnt X ℙ n t) {ω | a ≤ empiricalMean X n ω}
```

Appendix: Combining the Bounds

- ① **Log of limit to one:** If $f_n \rightarrow 1$ in $\mathbb{R}_{\geq 0}^\infty$, then $\log f_n \rightarrow 0$

```
private lemma ennreal_log_tendsto_zero_of_tendsto_one {f : ℕ → ℝ≥0∞}
  (h : Tendsto f atTop (ℕ 1)) :
  Tendsto (fun n => (f n).log) atTop (ℕ 0)
```

- ② **Inverse sequence vanishes in $\overline{\mathbb{R}}$:** $\frac{1}{n} \rightarrow 0$

```
private lemma ereal_inv_nat_tendsto_zero :
  Tendsto (fun n : ℕ => 1 / ((n : ℝ) : EReal)) atTop (ℕ 0)
```

- ③ **Product of vanishing sequences:** If $f_n \rightarrow 0$ and $g_n \rightarrow 0$ in $\overline{\mathbb{R}}$, then $f_n g_n \rightarrow 0$

```
private lemma ereal_mul_tendsto_zero_of_tendsto_zero_of_bounded
  {f g : ℕ → EReal} (hf : Tendsto f atTop (ℕ 0)) (hg : Tendsto g atTop (ℕ 0)) :
  Tendsto (fun n => f n * g n) atTop (ℕ 0)
```

- ④ **Typical event limit:** For $a < \mathbb{E}[X]$, $\mathbb{P}(\overline{X}_n \geq a) \rightarrow 1$ (via SLLN)

```
private lemma less_exp_imp_limit_prob_less_mean_one (a : ℝ) (h : a < E[X 0]) :
  Tendsto (fun n : ℕ => (ℙ {ω | a ≤ empiricalMean X n ω} : ENNReal)) atTop (ℕ 1)
```

- ⑤ **Cramér's theorem:** The empirical mean satisfies the LDP with rate function I

```
theorem cramers_theorem :
  LargeDeviationPrinciple (empiricalMean X) (upperTailRateFunction X)
```

Appendix: Probability Theory Foundations

```
-- Moment-generating function of a real random variable -/
def mgf (X :  $\Omega \rightarrow \mathbb{R}$ ) ( $\mu$  : Measure  $\Omega$ ) (t :  $\mathbb{R}$ ) :  $\mathbb{R}$  :=  $\mu$ [fun  $\omega \Rightarrow \exp (t * X \omega)$ ]

-- Cumulant-generating function -/
def cgf (X :  $\Omega \rightarrow \mathbb{R}$ ) ( $\mu$  : Measure  $\Omega$ ) (t :  $\mathbb{R}$ ) :  $\mathbb{R}$  := log (mgf X  $\mu$  t)

-- **Chernoff bound** on the upper tail -/
theorem measure_ge_le_exp_cgf [IsFiniteMeasure  $\mu$ ] ( $\varepsilon$  :  $\mathbb{R}$ ) (ht :  $0 \leq t$ )
  (h_int : Integrable (fun  $\omega \Rightarrow \exp (t * X \omega)$ )  $\mu$ ) :
     $\mu$ .real { $\omega \mid \varepsilon \leq X \omega$ }  $\leq \exp (-t * \varepsilon + \text{cgf } X \mu t)$ 

-- Exponentially tilted measure -/
def Measure.tilted ( $\mu$  : Measure  $\alpha$ ) (f :  $\alpha \rightarrow \mathbb{R}$ ) : Measure  $\alpha$  :=
   $\mu$ .withDensity (fun x  $\mapsto ((\exp (f x) / \int y, \exp (f y) \partial \mu : \mathbb{R}) : \mathbb{R}_{\geq 0}) : \alpha \rightarrow \mathbb{R}_{\geq 0 \infty})$ 

lemma isProbabilityMeasure_tilted [NeZero  $\mu$ ]
  (hf : Integrable (fun x  $\mapsto \exp (f x)$ )  $\mu$ ) :
    IsProbabilityMeasure ( $\mu$ .tilted f)

-- **Strong law of large numbers** (Etemadi) -/
theorem strong_law_ae_real (X :  $\mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}$ ) (hint : Integrable (X 0)  $\mu$ )
  (hindep : Pairwise (( $\cdot \perp_i [\mu] \cdot$ ) on X))
  (hident :  $\forall i$ , IdentDistrib (X i) (X 0)  $\mu \mu$ ) :
     $\forall^m \omega \partial \mu$ , Tendsto (fun n :  $\mathbb{N} \Rightarrow (\sum i \in \text{range } n, X i \omega) / n$ ) atTop ( $\mathcal{N} \mu [X 0]$ )
```

Appendix: Measure Theory & Analysis Foundations

```
-- A measure  $\mu$  is a probability measure if  $\mu(\Omega) = 1$  -/  
class IsProbabilityMeasure ( $\mu$  : Measure  $\alpha$ ) : Prop where  
  measure_univ :  $\mu$  univ = 1  
  
-- Integrable  $f$   $\mu$  means  $f$  is measurable and  $\int \|f\| \partial\mu < \infty$  -/  
def Integrable ( $f$  :  $\alpha \rightarrow \varepsilon$ ) ( $\mu$  : Measure  $\alpha$ ) : Prop :=  
  AESTronglyMeasurable  $f$   $\mu$   $\wedge$  HasFiniteIntegral  $f$   $\mu$   
  
-- The variance of  $X$  under the tilted measure is the 2nd derivative of the CGF -/  
lemma variance_tilted_mul (ht :  $t \in \text{interior}(\text{integrableExpSet } X \mu)$ ) :  
  Var[ $X$ ;  $\mu$ .tilted ( $t * X \cdot$ )] = iteratedDeriv 2 (cgf  $X \mu$ )  $t$   
  
-- Jensen's inequality for probability measures -/  
theorem ConvexOn.map_integral_le [IsProbabilityMeasure  $\mu$ ]  
  (hg : ConvexOn  $\mathbb{R}$   $s$   $g$ ) (hgc : ContinuousOn  $g$   $s$ ) (hsc : IsClosed  $s$ )  
  (hfs :  $\forall^m x \partial\mu, f \ x \in s$ ) (hfi : Integrable  $f$   $\mu$ )  
  (hgi : Integrable ( $g \circ f$ )  $\mu$ ) :  
   $g \left( \int x, f \ x \partial\mu \right) \leq \int x, g \ (f \ x) \partial\mu$ 
```

Appendix: Orders, Filters & Extended Reals

```
-- Tendsto f l1 l2: the generic "limit of a function" predicate -/
def Tendsto (f :  $\alpha \rightarrow \beta$ ) (l1 : Filter  $\alpha$ ) (l2 : Filter  $\beta$ ) := l1.map f ≤ l2

-- limsup of a function u along a filter f -/
def limsup (u :  $\beta \rightarrow \alpha$ ) (f : Filter  $\beta$ ) :  $\alpha$  := limsSup (map u f)

-- liminf of a function u along a filter f -/
def liminf (u :  $\beta \rightarrow \alpha$ ) (f : Filter  $\beta$ ) :  $\alpha$  := limsInf (map u f)

-- atTop: the filter representing the limit  $\rightarrow \infty$  -/
def atTop [Preorder  $\alpha$ ] : Filter  $\alpha$  := inf a,  $\mathcal{P}$  (Ici a)

lemma eventually_atTop : ( $\forall^f x$  in atTop, p x)  $\leftrightarrow$   $\exists$  a,  $\forall b \geq a$ , p b

-- The type of extended real numbers  $[-\infty, \infty]$  -/
def EReal := WithBot (WithTop  $\mathbb{R}$ )

-- The extended nonnegative real numbers  $[0, \infty]$  -/
def ENNReal := WithTop  $\mathbb{R}_{\geq 0}$ 

-- Logarithm on  $\mathbb{R}_{\geq 0\infty}$  to EReal:  $\log 0 = \perp$ ,  $\log \top = \top$  -/
noncomputable def ENNReal.log (x :  $\mathbb{R}_{\geq 0\infty}$ ) : EReal :=
  if x = 0 then  $\perp$  else if x =  $\top$  then  $\top$  else Real.log x.toReal
```